UTTKARSH RUPAREL

+91-9885000237 · rupareluttkarsh2309@gmail.com
LinkedIn · GitHub · Portfolio · LeetCode

Skills

Languages: Go, TypeScript, JavaScript, Python, Java, Rust
Backend & Systems: Node.js (Express, Fastify), Gin, FastAPI, REST, GraphQL, WebSockets, gRPC, Protocol Buffers, NGINX, Kong, PostgreSQL, MongoDB, Redis, Kafka, RabbitMQ, ClickHouse, Neo4j
Infrastructure & Observability: Docker, Kubernetes, AWS (MSK, ECR, SQS), Linux, Terraform, Helm, ArgoCD, CI/CD (GitHub Actions), Cloudflare, Prometheus, Grafana, Loki, OpenTelemetry, Jaeger
Frontend & Testing: React, Next.js, Astro, Tailwind CSS, Redux, D3.js, k6
Foundations: System Design, Distributed Systems, Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Database Management Systems

Projects

ESX — Stock Exchange Infrastructure
Go, Kafka, AWS (EKS, RDS, MSK), Redis, PostgreSQL, gRPC, Protocol Buffers, FIX Protocol 4.2, Docker, Kubernetes, Terraform, Jaeger, ArgoCD

- Engineered a production-grade securities exchange — 8 microservices (order gateway, settlement, matching engine, clearing house) over Kafka and gRPC, replicating core systems behind NASDAQ.
- Built a high-concurrency order gateway in Go sustaining 6,000+ RPS at sub-40ms latency via gRPC streaming and a Buffer & Drain pattern (500k-capacity buffered channels) to decouple ingestion from matching, surviving 300% load spikes.
- Implemented atomic DvP settlement with an idempotent double-entry ledger, preventing duplicate transactions via unique ID tracking and atomic PostgreSQL upserts — all movements net to zero.

Hatch — Self-hosted Cloud Deployment Platform
Go (Gin), AWS (ECS, ECR, Fargate, ALB), Terraform, RabbitMQ, Redis, PostgreSQL, Docker, Next.js

- Architected a self-hosted Render/Railway alternative on AWS — independent Go microservices for cloning, building and deploying containers on ECS Fargate, communicating through RabbitMQ with real-time log streaming via Redis pub/sub over WebSockets.
- Achieved 3,500 RPS at p99 < 75ms with zero failed requests across 5,000 requests (ApacheBench) and sustained 716 req/s at p95 < 9ms under 50 concurrent VUs (k6).
- Provisioned the full AWS stack via Terraform — VPC, subnets, security groups, ECS cluster, ALB with path-based routing per deployment — with a sqlc-generated type-safe PostgreSQL layer tracking deployment lifecycle across 5 states (queued → building → deploying → live → failed).

Sandboxed — Competitive Coding Contest Platform
Go, TypeScript, PostgreSQL, Redis, BullMQ, Socket.IO, Docker, AWS

- Architected a real-time coding contest platform across 3 independently deployed services — verdict pipeline flows from Go judge → Redis Pub/Sub → Node.js listener → WebSocket fan-out to all room participants in real time.
- Built a sandboxed code execution engine in Go — disposable Docker container per submission, network disabled, pids capped, 5 languages supported.
- Implemented a Redis-backed proctoring system with fullscreen enforcement, tab-switch detection, and a TTL-scoped violation counter that auto-kicks participants on the 4th breach.

Education

B.Tech in Computer Science & Engineering |	**CGPA: 7.53**
Symbiosis Institute of Technology, Pune	(2023 - 2027)
Higher Secondary Education |	**Percentage: 85.5%**
Excellencia Junior College, Secunderabad	(2021 - 2023)

Leadership & Achievements

Competitive Programming Co-Head, GDSC SIT Pune (July 2025 – Present) — Led coding contests, workshops, and mentorship sessions for 100+ students across the institute.